SMART UNIVERSITY ERP + LMS

Enterprise System Design & Implementation Roadmap

1. Project Identity & Scope

This project is an enterprise-level SaaS solution designed to digitize the complete academic and administrative lifecycle of a university. It combines a robust **ERP (Enterprise Resource Planning)** for administration and a comprehensive **LMS (Learning Management System)** for pedagogy.

2. Core Academic Structure

The system utilizes a hierarchical data model to ensure logical data isolation and scalability:

- **University:** Root entity (Multi-tenant support).
- **Faculty:** Grouping of related departments (e.g., Engineering, Business).
- **Department:** Administrative unit for specific academic programs.
- **Program & Semester:** Structured timelines for degree completion.
- **Courses & Credits:** The fundamental units of registration and CGPA calculation.

3. Role-Based Access Control (RBAC)

Role	Key Permissions & Access
Super Admin	SaaS level management, university onboarding, system logs.
University Admin	Faculty/Department setup, Staff recruitment, Financial oversight.
Faculty / Teacher	LMS management, attendance, exam creation, result publishing.
Advisor	Student course registration approval, academic counseling.

Role	Key Permissions & Access
Student	Course enrollment, payment, LMS access, transcript viewing.
Accounts Officer	Fee verification, scholarships, payroll, and invoice audits.

4. Major System Modules

Academic & Registration Engine

- Automated credit check and prerequisite verification.
- Add/Drop/Retake workflow with Advisor approval.

Examination & CGPA System

- MCQ, CQ, Viva, and Assignment support.
- Automatic Grade Point calculation and dynamic transcript generation.

Integrated LMS & Live Classes

- Content distribution (PDF, Video, Notes).
- API integration for Zoom/Google Meet with automated attendance tracking.

Financial Module

- Payment Gateway Integration: **SSLCommerz** **bKash** **Nagad**
- Real-time billing based on credit hours and scholarship waivers.

5. Recommended Technology Stack

> **Frontend:** Angular 19+ (Modular Architecture) & TailwindCSS
> **Backend:** Java Spring Boot, Spring Security, JWT Authentication
> **Database:** MySQL (Relational Model for consistent academic records)
> **DevOps:** Docker Containers, AWS or DigitalOcean hosting

6. Implementation Roadmap

1. **Phase 1:** Foundation (Auth, RBAC, Academic Hierarchy).

2. **Phase 2:** Student Life Cycle (Registration & Profiles).
3. **Phase 3:** Academic Operations (LMS & Exam Engine).
4. **Phase 4:** Financials & External Integrations (Payments/Video APIs).
5. **Phase 5:** Analytics (Dashboards for Revenue and Student Performance).
